

Prabhleen Kaur

+91 8448828689 | prabhleen26kaur@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Bharati Vidyapeeth College of Engineering, GGSIPU

Bachelor of Technology, Computer Science SGPA: 8.76

New Delhi

August 2024 – Present

Venkateshwar International School

Grade 12: 87.6

New Delhi

April 2022 - March 2023

Venkateshwar International School

Grade 10: 93.5

New Delhi

April 2020 - March 2021

TECHNICAL SKILLS

Languages: Java, C++, Python, SQL, JavaScript

Web Development: React, Node.js, Express, HTML, CSS

AI and Machine Learning: YOLOv8, DeepFace, OpenCV, NumPy, PyTorch, TensorFlow

Cloud and Databases: AWS (S3, DynamoDB, SNS)

Tools: GitHub, VS Code

RESEARCH

Adversarial Defense in Brain Tumor Detection | Kennesaw State University

April 2025 – Present

- Research Assistant under Dr. Sumit Chakravarty — built tumor detection pipeline on surgical hyperspectral images using HybridSN and SSRN; tested 5 adversarial attack types.
- Designing an autoencoder-based defense pipeline to detect and reconstruct adversarially corrupted images before inference.

PROJECTS

IRIS – Intelligent Real-Time Infraction Surveillance System

- Built a real-time surveillance system using YOLOv8 and InsightFace for gender/age inference across 3 detection modes, with cross-session repeat offender recognition using 512-d face embeddings persisted in DynamoDB.
- Designed an async AWS evidence pipeline triggering S3 uploads, DynamoDB logging, and SNS alerts in parallel threads on every violation, with auto-generated PDF reports and violation heatmaps.
- Engineered a multi-threaded architecture decoupling the CV detection loop, inference worker, and live Streamlit dashboard with sub-second metric updates and real-time crowd density alerts.

EduConnect – Full-Stack ERP System (SIH 2025)

- Built a production-grade full-stack ERP platform (React, Node.js/Express, Firebase) with scalable architecture, role-based access control, and real-time system design.
- Architected a modular ecosystem covering academics, finance, transport, and placement analytics using a hybrid data model.
- Delivered AI-powered automation (timetable generation, career guidance) and real-time services (messaging, complaint tracking), enabling end-to-end institutional digitization.

Smart Meter Data Analytics & Load Forecasting (EcoCode Hackathon)

- Developed an end-to-end energy analytics pipeline using a hybrid ensemble (LSTM, BiLSTM, TFT) and Groq API to deliver real-time, weather-aware load forecasting.
- Built a telemetry dashboard using React and PostgreSQL to visualize demand patterns and manage smart meter data.
- Automated MLOps workflows via GitHub Actions, implementing serverless scheduling for model retraining to optimize system scalability and operational costs.